1 Introduction

$$G = G_0 + G_0 V G = G_0 + G V G_0 \tag{1}$$
$$T = V + V G_0 T = V + T G_0 V \tag{2}$$
$$G = G_0 + G_0 T G_0 \tag{3}$$

For

$$V = \hat{V}\delta(r) \tag{4}$$

The real-space Green's function is given by

$$G(r, r', z) = G_0(r - r', z) + \int dr'' G_0(r - r'', z) V(r'') G(r'', r', z) \tag{5}$$
$$= G_0(r - r', z) + G_0(r, z)\hat{V} G(0, r', z). \tag{6}$$

Solving for the T-matrix,

$$T(z) = \left(1 - \hat{V} G_0(r = 0, z)\right)^{-1} \hat{V} \tag{7}$$
$$= \hat{V}\left(1 - G_0(r = 0, z)\hat{V}\right)^{-1}. \tag{8}$$

The full real-space Green's function is therefore

$$G(r, r', z) = G_0(r - r', z) + G_0(r, z) T(z) G_0(-r', z). \tag{9}$$

To get the retarded or advanced Green's function, we substitute $z = \omega \pm i\delta$.

$$G^{R(A)}(r, r', \omega) = G(r, r', \omega \pm i\delta). \tag{10}$$

The local density of states is given by

$$\rho(r, \omega) = -\frac{G^R(r, r, \omega) - G^A(r, r, \omega)}{2\pi i} \tag{11}$$
$$= \rho_0(\omega) - \frac{1}{2\pi i}\left(G_0^R(r, \omega) T^R(\omega) G_0^R(-r, \omega) - G_0^A(r, \omega) T^A(\omega) G_0^A(-r, \omega)\right) \tag{12}$$

The unperturbed real-space Green's functions can be calculated from the k-space Green's functions,

$$G_0(r, z) = \int \frac{d^2k}{(2\pi)^2} g_0(k, z) e^{ikr}, \tag{13}$$

and the unperturbed LDOS is given by

$$\rho_0(\omega) = -\frac{G_0^R(0, \omega) - G_0^A(0, \omega)}{2\pi i} \tag{14}$$
$$= -\frac{1}{2\pi i}\int \frac{d^2k}{(2\pi)^2}\left(g_0^R(k, \omega) - g_0^A(k, \omega)\right) \tag{15}$$
$$= \int \frac{d^2k}{(2\pi)^2} A(k, \omega), \tag{16}$$

where

$$A(k, \omega) = -\frac{g_0^R(k, \omega) - g_0^A(k, \omega)}{2\pi i}. \tag{17}$$

Next, we define the FT-LDOS as

$$\rho(q, \omega) = \int dr \rho(r, \omega) e^{-iqr} \tag{18}$$
$$= \rho_0(\omega)\delta(q) - \frac{1}{2\pi i}\int dr \frac{d^2k}{(2\pi)^2}\frac{d^2k'}{(2\pi)^2}\left(g_0^R(k, \omega) T^R(\omega) g_0^R(k', \omega) - g_0^A(k, \omega) T^A(\omega) g_0^A(k', \omega)\right) e^{i(k-k'-q)r} \tag{19}$$
$$= \rho_0(\omega)\delta(q) - \frac{1}{2\pi i}\int \frac{d^2k}{(2\pi)^2}\left(g_0^R(k, \omega) T^R(\omega) g_0^R(k - q, \omega) - g_0^A(k, \omega) T^A(\omega) g_0^A(k - q, \omega)\right) \tag{20}$$

Let

$$S(q,\omega) = \int \frac{d^2k}{(2\pi)^2} g_0^R(k,\omega) T^R(\omega) g_0^R(k-q,\omega) \tag{21}$$

$$S^\dagger(q,\omega) = \int \frac{d^2k}{(2\pi)^2} g_0^A(k-q,\omega) T^A(\omega) g_0^A(k,\omega) \tag{22}$$

By shifting the integration variable $k \to k+q$, we get

$$S^\dagger(q,\omega) = \int \frac{d^2k}{(2\pi)^2} g_0^A(k,\omega) T^A(\omega) g_0^A(k+q,\omega). \tag{23}$$

Therefore,

$$\rho(q,\omega) = \rho_0(\omega)\delta(q) - \frac{S(q,\omega) - S^\dagger(-q,\omega)}{2\pi i}. \tag{24}$$

The last term is usually what corresponds to the QPI.

1.1 Calculation of S

$$\begin{aligned} S(q,\omega) &= \int \frac{d^2k}{(2\pi)^2} g_0^R(k,\omega) T^R(\omega) g_0^R(k-q,\omega) && (25)\\ &= \int \frac{d^2k}{(2\pi)^2} d^2r d^2r' G_0^R(r,\omega) T^R(\omega) G_0^R(r',\omega) e^{-ikr-ikr'+iqr'} && (26)\\ &= \int d^2r G_0^R(r,\omega) T^R(\omega) G_0^R(-r,\omega) e^{-iqr} && (27) \end{aligned}$$

Let

$$S(r,\omega) = G_0^R(r,\omega) T^R(\omega) G_0^R(-r,\omega). \tag{28}$$

Note that $S(r,\omega) = G^R(r,r,\omega) - G_0^R(r=0,\omega)$. Then

$$S(q,\omega) = \int d^2r S(r,\omega) e^{-iqr} \tag{29}$$

I.e., to get $S(q,\omega)$, we FT $S(r,\omega)$. $S(r,\omega)$ is calculated from real-space Green's functions, which are calculated by IFT-ing $g_0^R(k,\omega)$ (Eq. (13)).

1.2 Symmetries of H

$$\begin{aligned} H_k &= U_T H_{-k}^* U_T^\dagger && (30)\\ H_k &= -U_C H_{-k}^* U_C^\dagger && (31)\\ H_k &= -U_S H_k U_S^\dagger && (32) \end{aligned}$$

1.3 Symmetries of G

$$\begin{aligned} g_0(k,z) &= U_T g_0^*(-k,z^*) U_T^\dagger && (33)\\ g_0(k,z) &= -U_C g_0^*(-k,-z^*) U_C^\dagger && (34)\\ g_0(k,z) &= -U_S g_0(k,-z) U_S^\dagger && (35) \end{aligned}$$

$$\begin{aligned} G_0(r,z) &= \int \frac{d^2k}{(2\pi)^2} g_0(k,z) e^{ikr} && (36)\\ &= U_T G_0^*(r,z^*) U_T^\dagger && (37)\\ &= -U_C G_0^*(r,-z^*) U_C^\dagger && (38)\\ &= -U_S G_0(r,-z) U_S^\dagger && (39) \end{aligned}$$

$$\begin{aligned} G_0^R(r,\omega) &= U_T G_0^A(r,\omega)^* U_T^\dagger && (40)\\ &= -U_C G_0^R(r,-\omega)^* U_C^\dagger && (41)\\ &= -U_S G_0^A(r,-\omega) U_S^\dagger && (42) \end{aligned}$$

1.4 Symmetries of T

$$\left(T^R(\omega)\right)^\dagger = \hat{V}^\dagger \left(1 - G_0^R(0,\omega)^\dagger \hat{V}^\dagger\right)^{-1} \tag{43}$$

$$= \hat{V} \left(1 - G_0^A(0,\omega)\hat{V}\right)^{-1} \tag{44}$$

$$= T^A(\omega) \tag{45}$$